Relative & Expected Frequency

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Relative frequency

What is relative frequency?

- **Relative frequency** is an **estimate of a probability** using results from an **experiment**
  - For a certain number of trials of that experience, the probability of 'success' is:

$$\frac{\text{Number of successful outcomes}}{\text{Total number of trials}}$$

- If you flip an unfair coin 50 times and it lands on heads 20 times, an **estimate** for the probability of the coin landing on heads is $\frac{20}{50}$ (its relative frequency)
  - That is the **best estimate** we can make, given the data we have
  - We do **not** know the **actual** probability
- The **more trials** that are carried out, the **more accurate** relative frequency becomes
  - It gets closer and closer to the **actual** probability

When will I be asked to use relative frequency?

- Relative frequency is used when **theoretical probabilities** are **unavailable** (or **not possible to calculate**)
  - For example, if you do not know the theoretical probability of being left-handed, you can run an experiment to find an estimate (the relative frequency)
- Sometimes theoretical probabilities are known
  - The **theoretical** probability of a **fair** coin landing on heads is 0.5
  - The **theoretical** probability of a **fair** standard six-sided dice landing on a six is $\frac{1}{6}$
- **Relative frequency** can be **compared** to a **theoretical probability** to test if a situation is **fair** or **biased**
  - If 100 flips of the coin give a relative frequency of 0.48 for landing on heads, the coin is likely to be fair

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

for landing on heads, the coin is likely to be biased (not fair)

What else do I need to know about relative frequency?

- Relative frequency assumes that there is an **equal chance of success** on each trial
  - The trials are independent of each other
    - For example, if choosing something out of a bag (a ball, or marble etc), it would need to be **replaced each time** to use relative frequency
- Any experiments used to calculate relative frequency should be **random**
  - If the experiment is not random, this could introduce **bias**

Examiner Tips and Tricks

If you have to choose the best estimate, choose the one with the most trials.

Worked Example

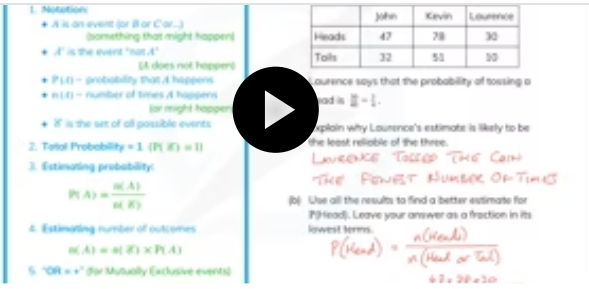
There are an unknown number of different coloured buttons in a bag.
Johan selects a button at random, notes its colour and replaces the button in the bag.
Repeating this 30 times, Johan notes that on 18 occasions he selected a red button.

Use Johan's results to estimate the probability that a button drawn at random from the bag is red.

Answer:

Taking 'red' to be a success, Johan had 18 successes out of a total of 30 trials.

$$P(\text{red}) = \frac{18}{30} = \frac{3}{5}$$



Did this video help you? Yes No

Expected frequency

What is expected frequency?

- **Expected frequency** refers to the **number of times** you would **expect** a particular outcome to occur
- It is found by **multiplying** the **probability** by the **number of trials**
  - If you flip a fair coin 100 times, you would expect 0.5 × 100 = 50 heads
- Sometimes you need to calculate the **relative frequency** first
  - If you flip a biased coin 40 times and get 10 heads, how many heads would you expect when flipping 100 times?
    - The relative frequency is $\frac{10}{40}$ = 0.25 from the first experiment
    - 0.25 × 100 = 25, you would expect to get heads 25 times from 100 throws

Examiner Tips and Tricks

Exam questions will not necessarily use the phrase "expected frequency", but might ask how many you "would expect".

Worked Example

There are 6 blue, 4 red and 5 yellow counters in a bag.
One counter is drawn at random and its colour noted.
The counter is then returned to the bag.

(a) Find the probability that a counter drawn from the bag is yellow.

Answer:

$$P(\text{Yellow}) = \frac{5}{15} = \frac{1}{3}$$

(b) How many times would you expect a yellow counter to be drawn, if this experiment is repeated 300 times?

Answer:

This is expected frequency so multiply the number of trials by the probability from part (a)

$$300 \times \frac{1}{3} = 100$$

We would expect 100 yellow counters

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